

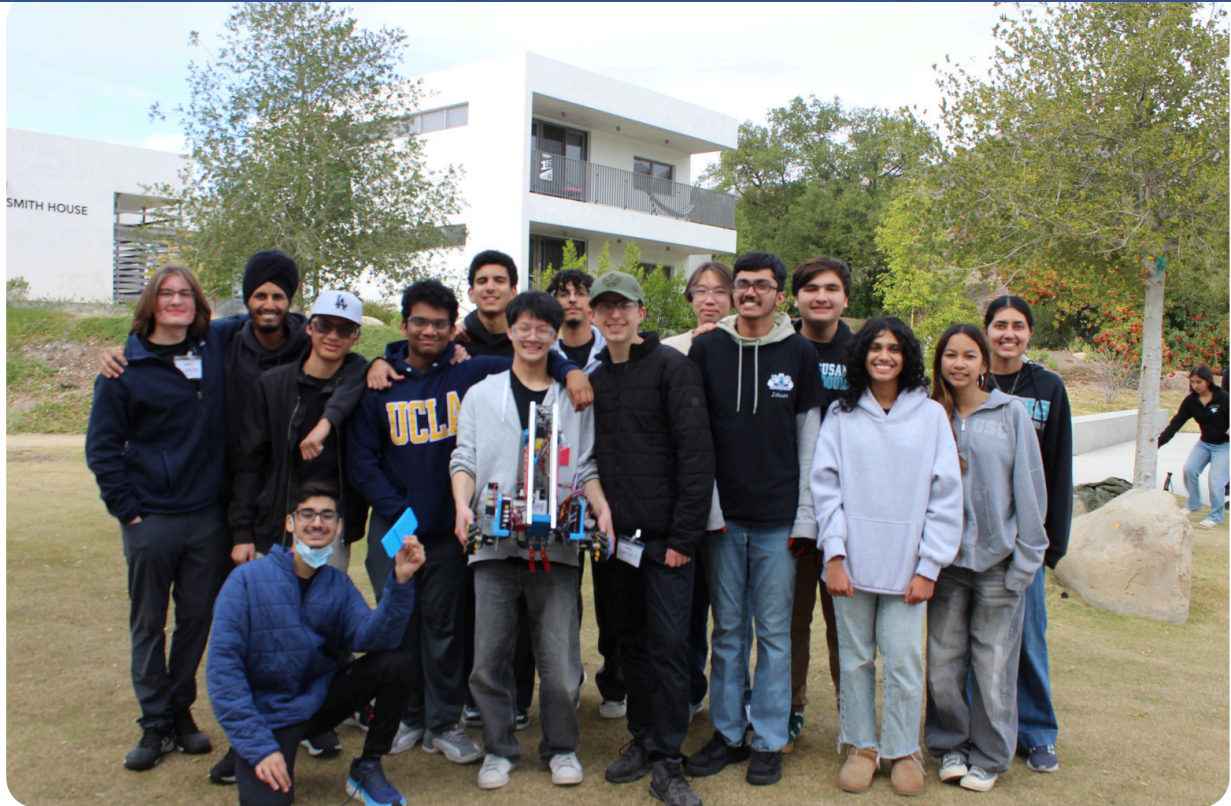
TEAM 20381



SANTA SUSANA HIGH SCHOOL

ADVISOR: BRUCE NEBENS

MEET OUR TEAM



**TEAM
CAPTAIN**

JONAS TIEU



**VICE
CAPTAIN**

AADI NAIR



OUTREACH LEAD
JOSH RODRIGUEZ

CHLOE ASIDERA
REESEHUERTA CAMPOS
AIDEN LONG
DANIEL ZAKI
KASRA HOOR



SOFTWARE LEAD
PRANAV BIJITH

ADITHI MURALIDAS
DILVEER GILL
LUCA FLOOD
NATHAN WIDMAN
WILLIAM OLSEN



MECHANICAL LEAD
GAVIN HUDSON

HARPREET GILL
DANIEL ZAKI
ISHAAN SHARMA
KASRA HOOR
MADISON LO
REESE HUERTA CAMPOS
MUHADDAS SADIQAIN

OUR MENTORS

WE do the work, but THEY keep us motivated and organized throughout the season



Tim Gray is the Vice President of Space Vector Corporation, an aerospace engineering company. They mainly produce lithium batteries for companies such as SpaceX and governmental organizations such as NASA and the Army. His sponsorship provided crucial financial support, helping our team stay on track as we aimed to elevate our robot's performance.



Vivian Abdallah was one of the first people we contacted and is now a valued mentor. As a VEX Robotics advisor, she expanded our experience beyond FIRST. She keeps us on top of deadlines, event logistics, and portfolio development while also strengthening our outreach. Her guidance has been invaluable in refining our team and increasing our impact.

GOAL #1: Improve Knowledge Transfer

In previous years, our performance has been determined by singular "talents"; we want to ensure our success without a single member solely contributing.

- Comprehensive Member Training Programs
- Team-Building Initiatives

GOAL #2: Strengthen Community Presence

We hoped to continue improving our community by engaging our growing team.

- Increased Involvement within Community Events
- Forging More Local Partnerships

GOAL #3: Utilize Odometry & Computer Vision in Autonomous

We previously neglected incorporating a sophisticated autonomous into our game strategy in favor of driver control; we realized this fault and pivoted to using odometry.

- Road Runner Research
- Implementation of OpenCV for a more precise intake

GOAL #4: Improve Lift Capabilities

To have more complex systems interact with our lift and achieve ascent + high-basket, we wanted to have a smoother lift system.

- Misumi Slides, Horizontal Extension
- Motor-Powered dual up-and-down string system

LIFT PROCESS

PLAN

MISUMI BASED



MEET 0

VERTICAL

MISUMI-SLIDE LIFT

BASE-MOUNTED

SPOOL MOTORS



MEET 1

STRING -> KEVLAR

MOTOR MOUNTING SHIFTED



MEET 2-3

1 -> 2 MISUMI SLIDES

SETS

INCREASED EXPANSION

STRONGER KEVLAR

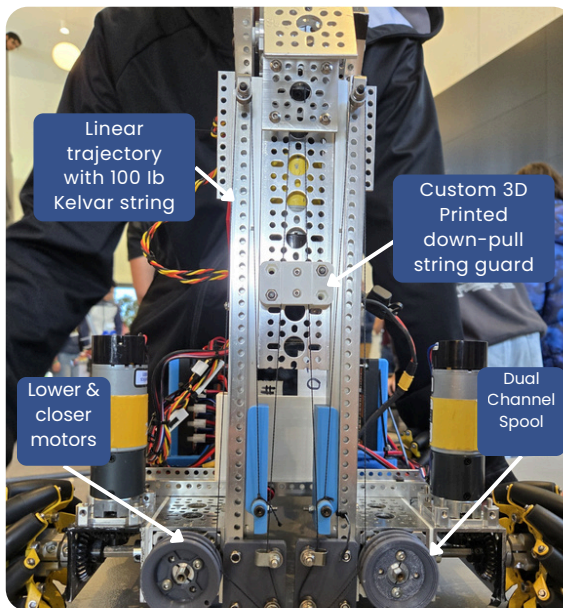
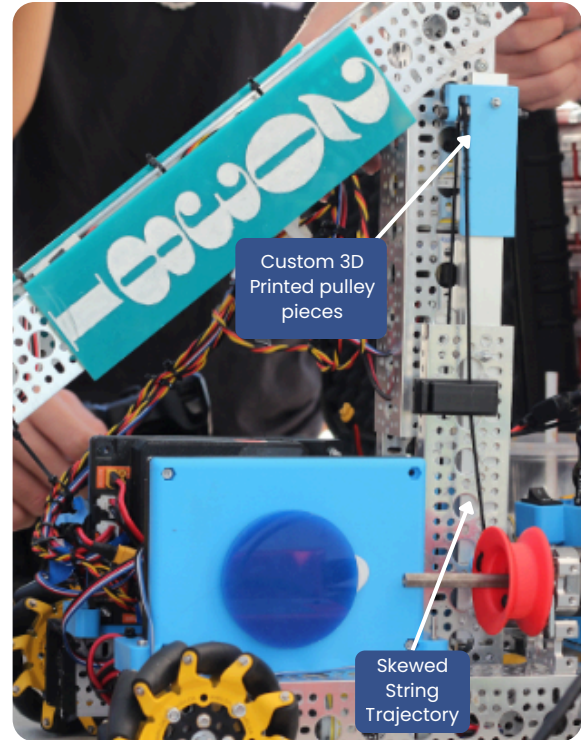
TENT STRING TENSIONER

Initial Lift Plan:

- **Reach high basket** with two-stage Misumi slides
- Insert a **dual C-channel "arm"** powered by a high torque motor

Meet 0 Design Flaws:

- **Basic paracord string frayed from friction** due to skewed trajectory as a result of motor placement
- **Lack of a down-pull string** prevented the ability to hang
- One set of Misumi slides made it **difficult to reach the high basket**

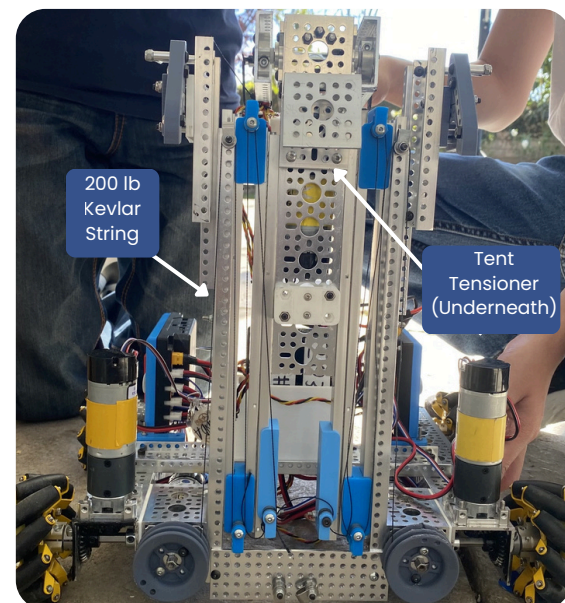


Meet 1 Design Flaws:

- Down-pull **strings untying** from the top anchor
- **Unable to reach inside the submersible** for extra samples as Misumi slides only expanded vertically
- **Spools exposed** in the back.

Meet 2-3 Design Flaws:

- **Still unable** to reach inside the submersible because **expansion was still vertical**
- 3rd level ascent was unachievable even with added Misumi slides, due to **center of mass balance issues**
- Spools still exposed from the back



PRE ILT

HIGH TORQUE MOTORS

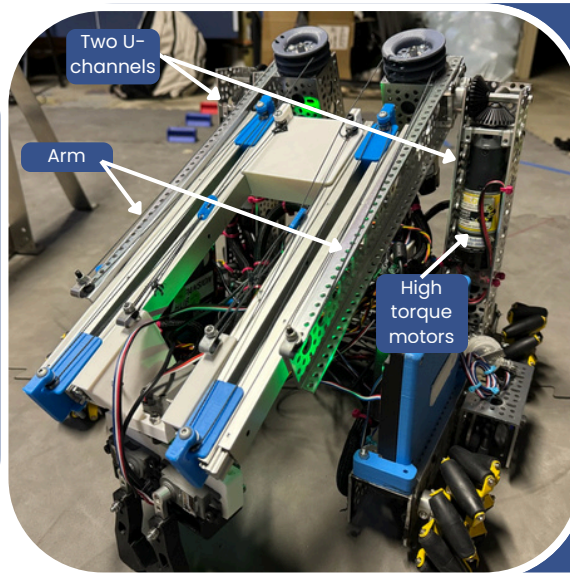
POWERED ARM ROTATION

ALLOWING FOR REACHING

HIGH BASKET VERTICALLY &

SUBMERSIBLE

HORIZONTALLY



LIFT TAKEAWAYS

- Further reinforcement or weight reduction is needed for arm stability
- Utilizing Kevlar string is the optimal material
- Encoder limits to stop overshooting, so string doesn't go astray
- Connecting multiple Misumi slides with custom 3D prints

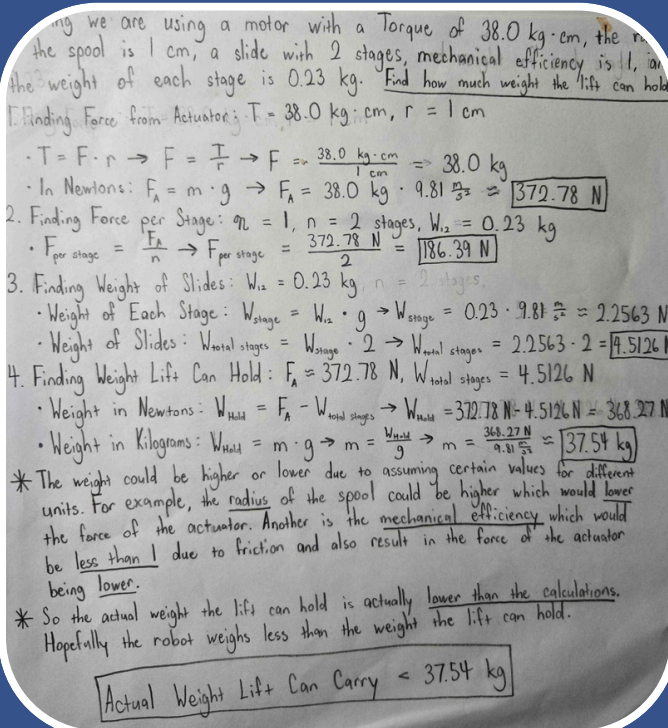
LIFT HIGHLIGHTS

Initial Lift Calculations

We were initially concerned with how much weight our lift could hold when trying to achieve 3rd level ascent, thus we decided to calculate it. Through the use of AP Physics I knowledge, finding this was possible.

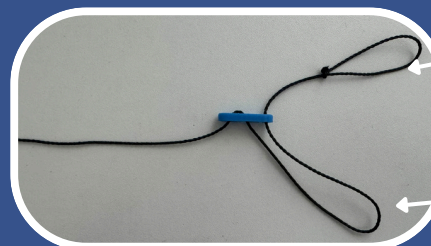
1. Using the torque equation, we must convert the torque into units of force since our lift is being powered by a motor
2. We must find the weight per stage of the lift when going up.
3. Convert the weight of the slides into Newtons
4. Finally, we found the actual weight the lift can carry by subtracting the force from the motor by the total weight of the slides.

It needs to be noted that the radius of the spool, mechanical efficiency, and gear ratio of the motor we decide to use can affect the results.



What is a tent tensioner?

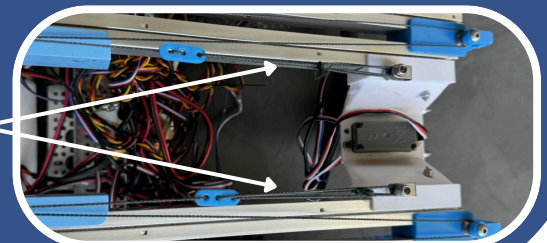
To address string tensioning, we developed and 3D printed an innovative mechanism **inspired by team 14525 TerraBats's Powerplay robot**. By incorporating tent tensioners and securing the string with a **common camping bowline knot**, we ensured **constant tension** at all times. A **simple tug down** on the tensioner was all it took to keep the string tight.



Bowline Knot at the End

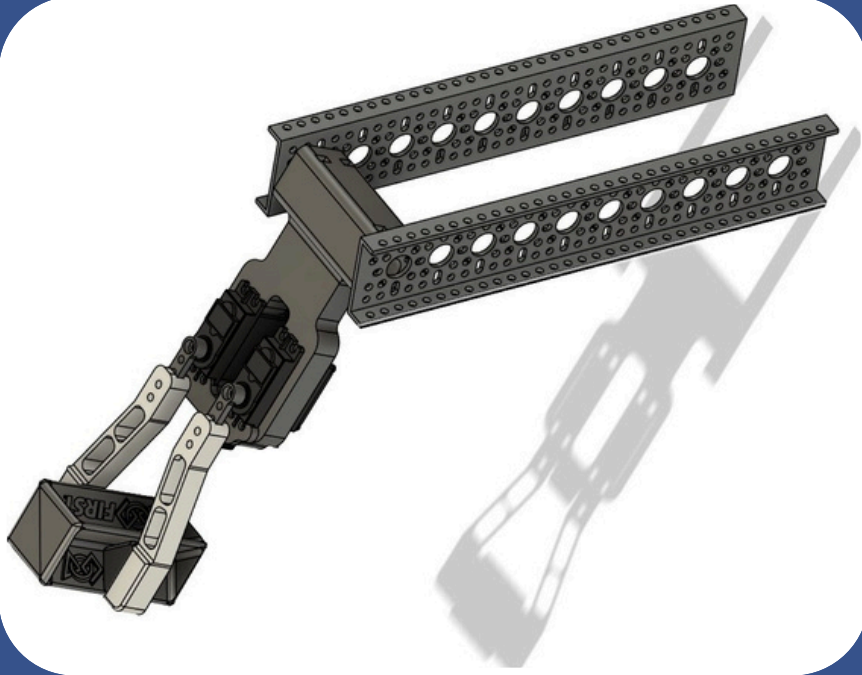
Loop from one side pulled out

Completely taut down-pull strings



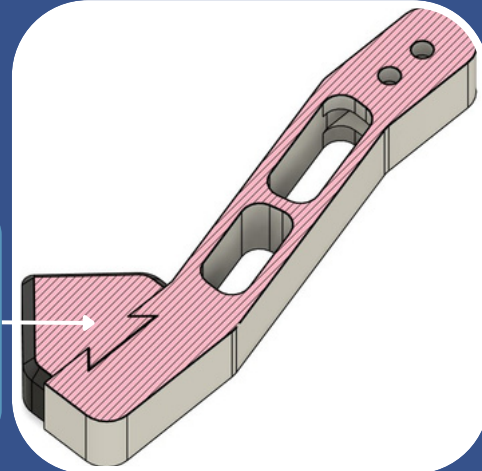
INTAKE PROCESS

MEET 0-3

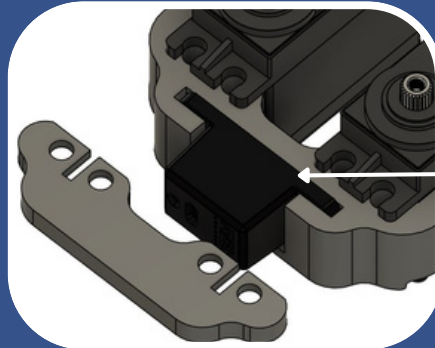


Above was our primary intake system during the season. The 3D printed claw arms have **custom TPU Inserts** to introduce a squishy grip onto specimens.

TPU inserts
were
placed
during
printing

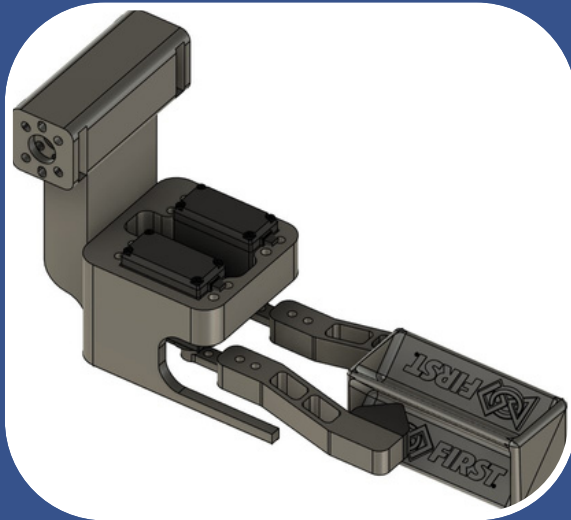


Proximity
sensor
added for
Meet 3 to
auto close
claw

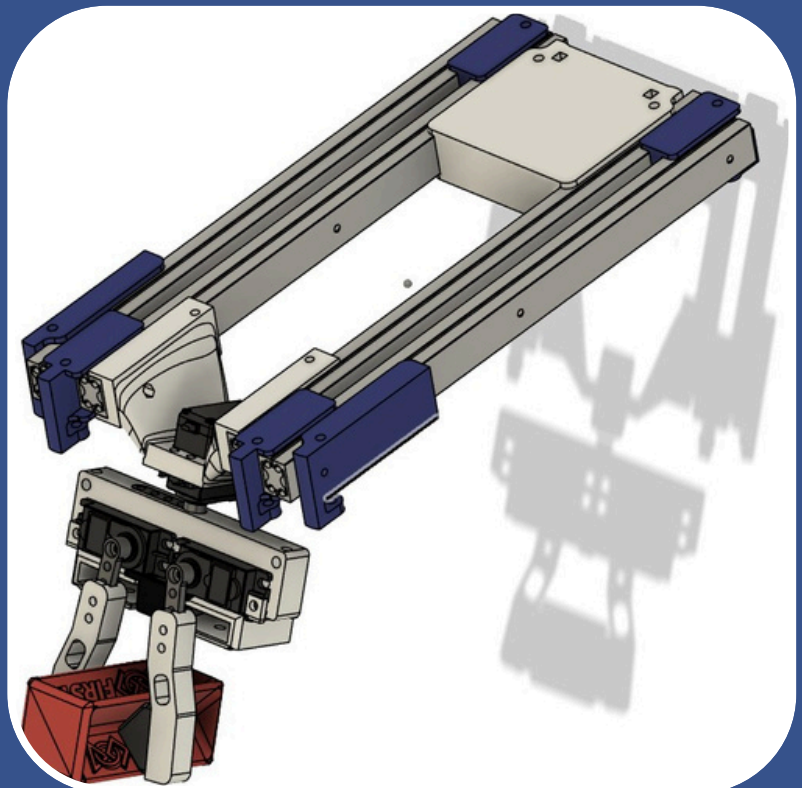


To the left is a past take on our initial claw design. Although a horizontally oriented claw would make grabbing samples off the ground easier, **fetching blocks in the submersible** would be **difficult**.

PRE ILT

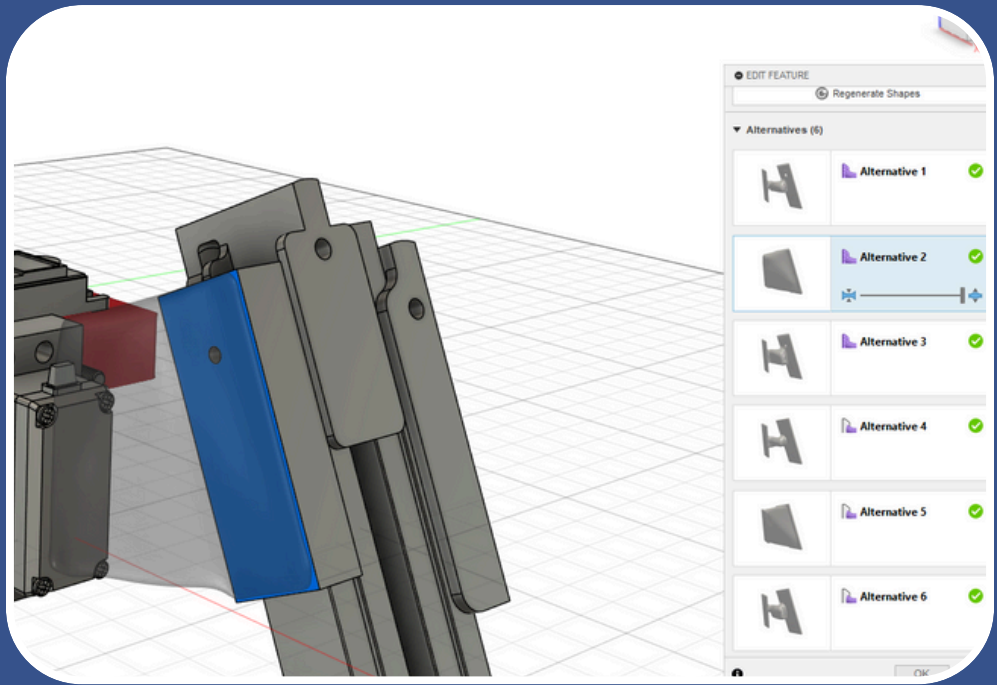


On the right is our ILT intake system. The mount connects to Misumi slides, utilizing **AI modeling**. Looking downwards on the claw is a **Logitech C920x camera** and **proximity sensor**. The claw **rotates along the yaw axis** through a servo, making it possible to **auto align** with samples using **OpenCV camera detection**.

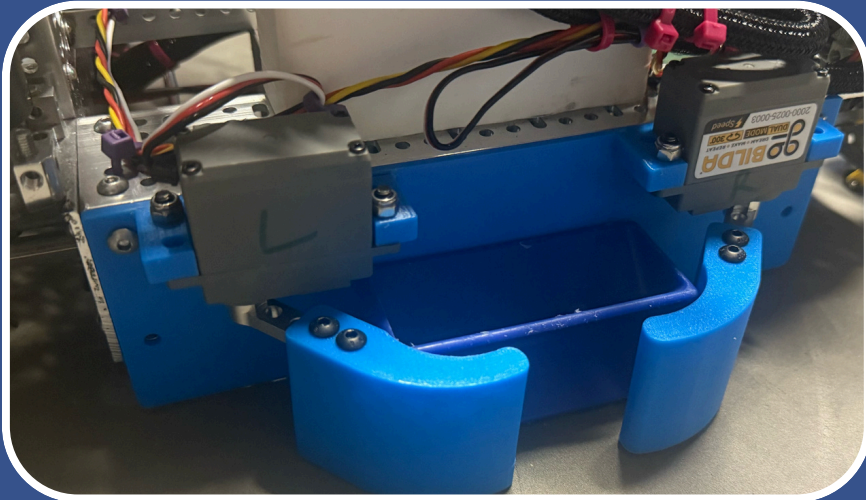


Ai Modeling

Using Fusion 360 Automated AI Modeling, we were able to select **desired faces to connect** and certain bodies to avoid. From there, we sorted through **generated iterations**, and decided on our final design. It allowed us to **simply create geometry**, and have Fusion 360 design difficult modeling for us, saving time and making a stronger part. This tool has been incredibly beneficial and something we will continue to utilize in the future.



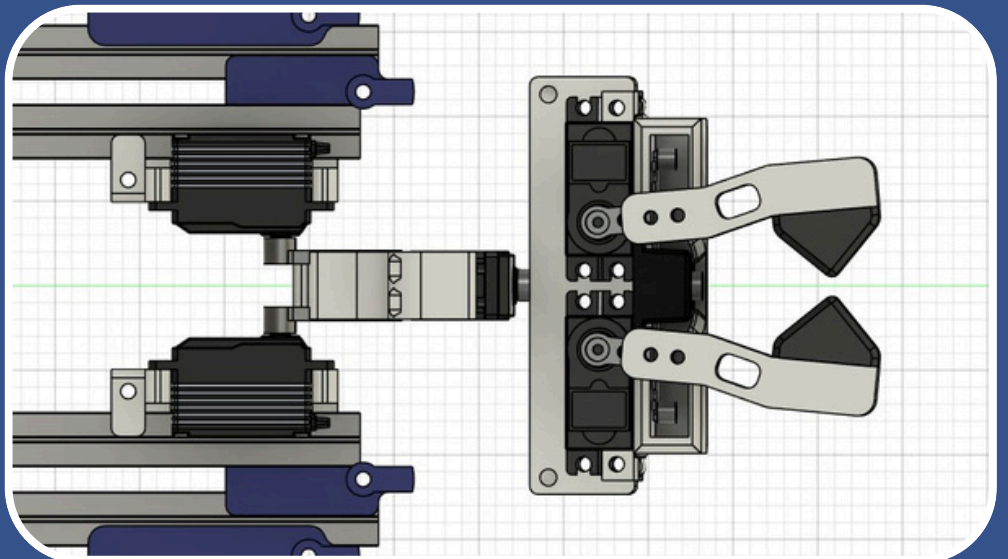
CAD HIGHLIGHTS



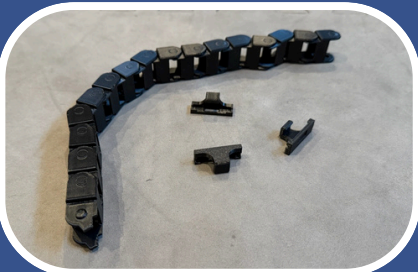
Rear Intake

To the left is our rear intake design, which plays a crucial role in our autonomous and tele-op modes. The claw is designed to quickly and efficiently collect blocks **from the ground, minimizing the time spent in picking up blocks with our primary intake**, allowing for a smoother and faster transition between tasks.

SoCal Intake



The addition of a **wrist** to allow for **more efficient autonomous movements**. Implementation of **Axon servos** for their additional torque.



Modified wire management models from online



Wheel armor to prevent sample blockage to movement

SUMMER WORKSHOP CAD VIDEOS

One thing we focused on during our summer workshops for incoming introductory students was teaching Fusion 360. Our experienced CAD members created videos to pass knowledge on to younger members. Projects provided hands-on experience for them to apply our teachings.

Week 1: Sketches

The foundation of Fusion 360 are sketches, **2D profiles** to be later expressed into 3D

Sketches Project

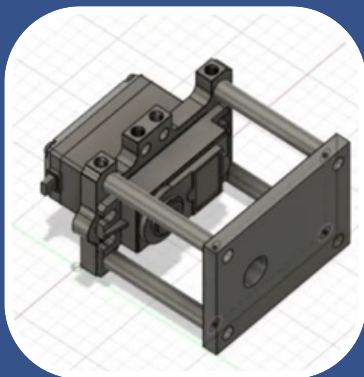
Importation of **goBILDA step files** parts, & integrating them in a **simple servo mount design**

Week 2: Build Tab

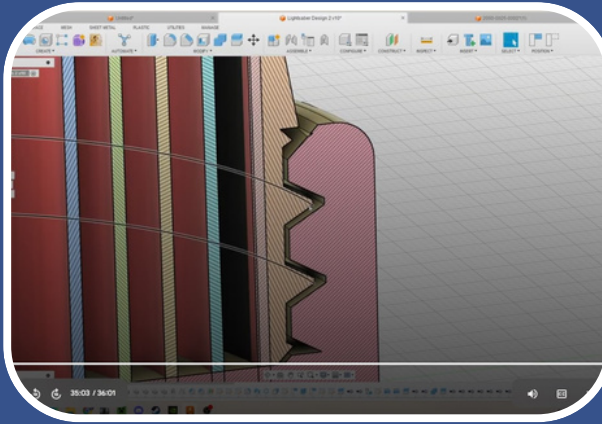
Members learned various **Fusion 360 features** offers to turn **2D sketches into 3D objects**

Build Tab Project

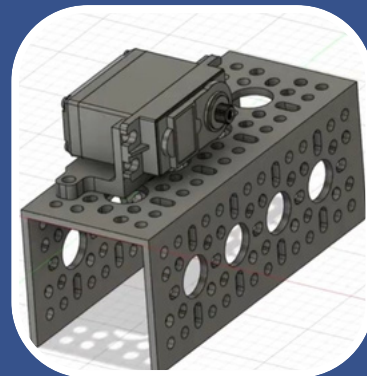
Modeling and learning a more **versatile servo mount design** process is **crucial** to many **claw oriented intakes**



Sketches Project



Build Tab lesson video



Build Tab Project

Week 5: Tips and Tricks

From integrating tolerances to section analysis, a **higher understanding was obtained**

Modify Project

Utilizing the modify tab, members created **Misumi slide pulley pieces**

Week 4: Modify

Improvement of 3D models in order to add **more functionality and complexity**

Week 3: Constraints

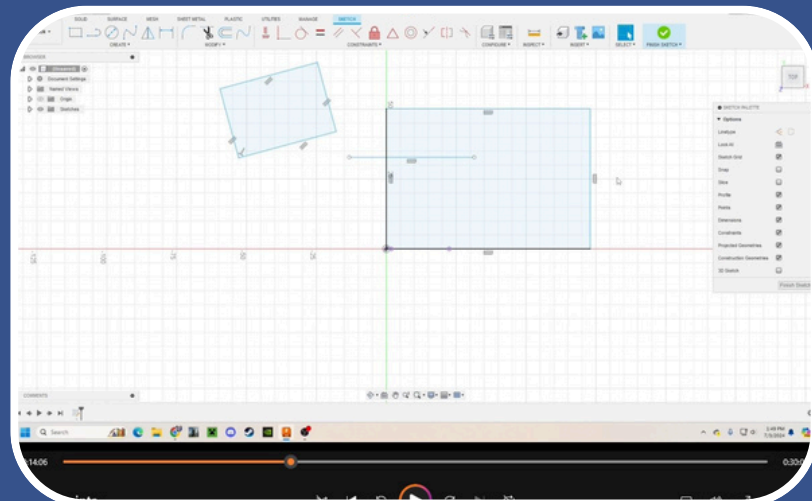
A vital aspect of Fusion 360 is to correctly make **parametric models** for **future iterations**

Final Project

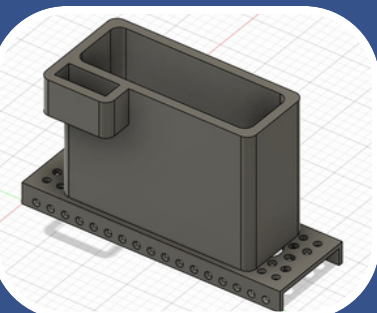
Using **everything learned**, members modeled their **own battery holder**



Modify Project

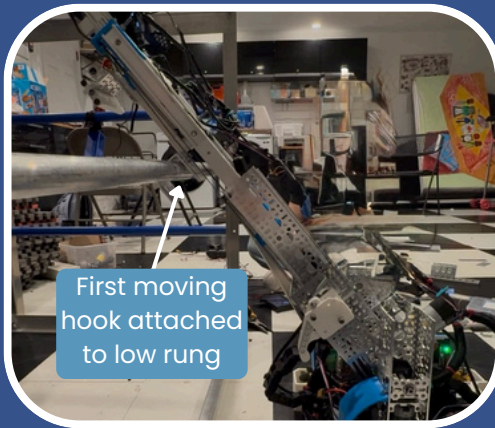


Our Fusion 360 lesson on teaching the basics of constraints in sketches

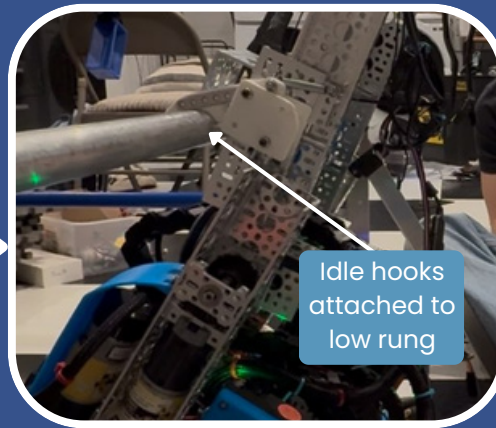


LEVEL 3 ASCENT PROCESS

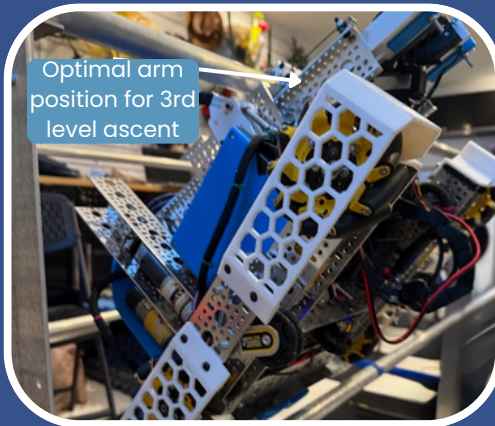
We designed "idle" or stationary hooks that enabled us to get 3rd level ascent. These idle hooks pair with a "moving" hook on our slides to allow for a climbing movement. The idle hook rotates and stops on a physical limit when hanging; a spring allows for it to flip down when climbing, and flip back up when hanging.



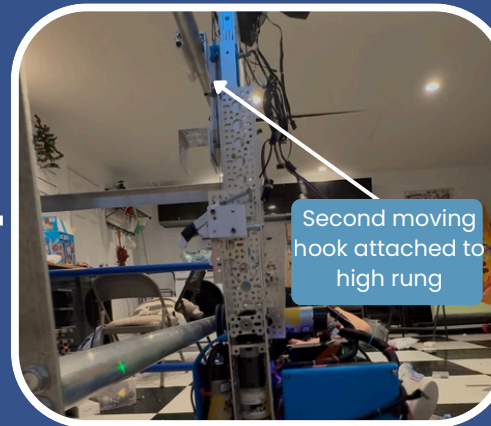
First moving hook attached to low rung



Idle hooks attached to low rung



Optimal arm position for 3rd level ascent



Second moving hook attached to high rung

STEP 1

LIFT EXTENDS BETWEEN LOW AND HIGH RUNG FOR THE FIRST MOVING HOOK TO LATCH TO LOW RUNG

STEP 2

LIFT DESCENDS SO IDLE HOOKS CAN FLAP AND ATTACH TO LOW RUNG

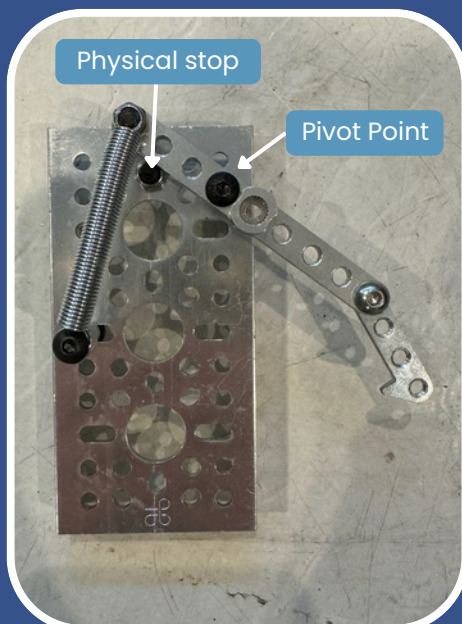
STEP 3

LIFT EXTENDS ONCE AGAIN TO ATTACH 2ND MOVING HOOK ONTO HIGH RUNG

STEP 4

LIFT DESCENDS FOR ROBOT TO GET OVER LOW RUNG AND ARM ADJUSTS TO PREVENT CONTACT WITH LOW RUNG

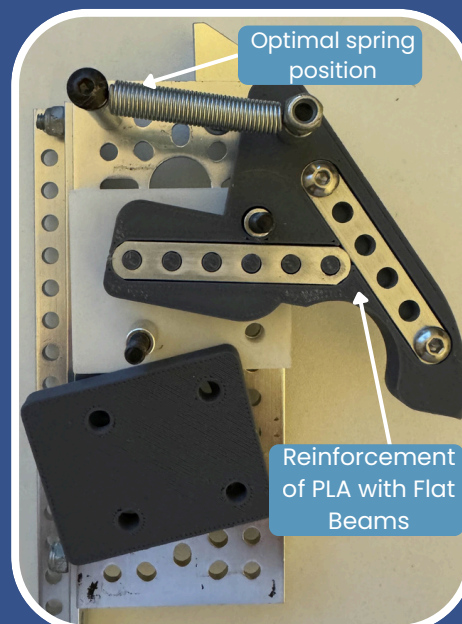
"Idle" Hooks



Physical stop

Pivot Point

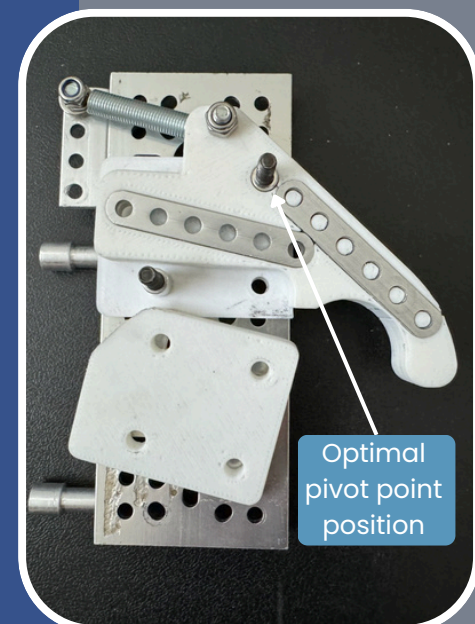
Original Idle hook design, equipped with **basic parts**, lacking **custom 3D prints**.



Optimal spring position

Reinforcement of PLA with Flat Beams

Idle hook design #2, implemented **custom 3D printed reinforcements**, also found **optimal spring position**



Optimal pivot point position

Finalized design, **changed pivot point** to make **hook flush with c-channel** when flipped down

DRIVER CONTROL



Our driver control system utilizes holonomic drive and includes:

- An **additive vector system** that strafes/rotates while counteracting friction
- “Fast” and “slow” controllers for refined movement

Proxy Sensors:

It was tedious to **adjust the specific height** of the arm to grab specimens, so we implemented a proxy sensor that automatically closes when it detects a specimen within **30 mm** of it.

Lift Vibrations

Originally, our lift extension was structured into **small intervals**, leading to large vibrations. We alleviated this by **increasing the interval lengths** accordingly.

DRIVE MOTOR EQUATIONS

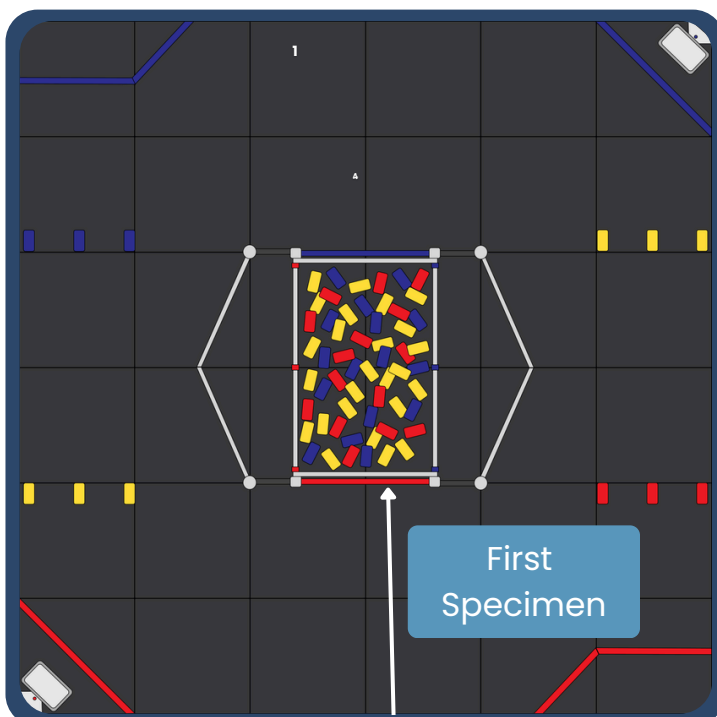
$$frontLeftPower = \frac{(y + x + rx)}{\max(|y| + |x| + |rx|, 1)}$$

$$backLeftPower = \frac{(y - x + rx)}{\max(|y| + |x| + |rx|, 1)}$$

$$backRightPower = \frac{(y + x - rx)}{\max(|y| + |x| + |rx|, 1)}$$

$$frontRightPower = \frac{(y - x - rx)}{\max(|y| + |x| + |rx|, 1)}$$

AUTONOMOUS (MEETS 0-2)



Our initial autonomous program utilized **multiple encoders** to measure the rotation of our motors and translate it to movements on their field and lift **extension/rotation**.

A flaw we realized is that encoders only support **concrete, sequential movements** rather than smooth curves, a problem only worsened by **voltage fluctuations** and **miter gear friction**.

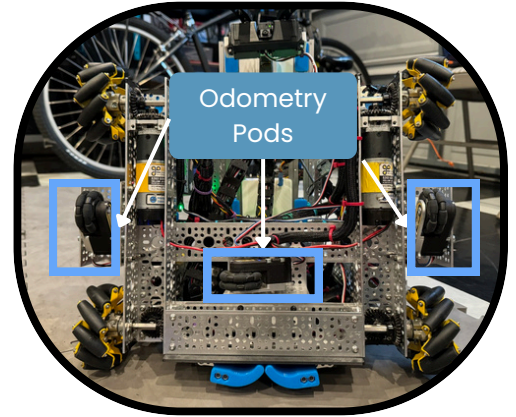
Because of this, we were only able to **clip one specimen on the submersible consistently**, as seen on the left.

AUTONOMOUS (MEETS 3+)

During the month-long break between Meets 2 and 3, we revamped our autonomous to utilize **odometry via Roadrunner**.

Our team made the decision to use **three odometry wheels** on the sides and middle of our robot to cover all **degrees of freedom without IMU**.

Prior to Meet 3, we utilized the **Trajectory Sequence** builder commands within Roadrunner to construct our paths and splines. This led to **issues with robot-field localization and manual measurements**.



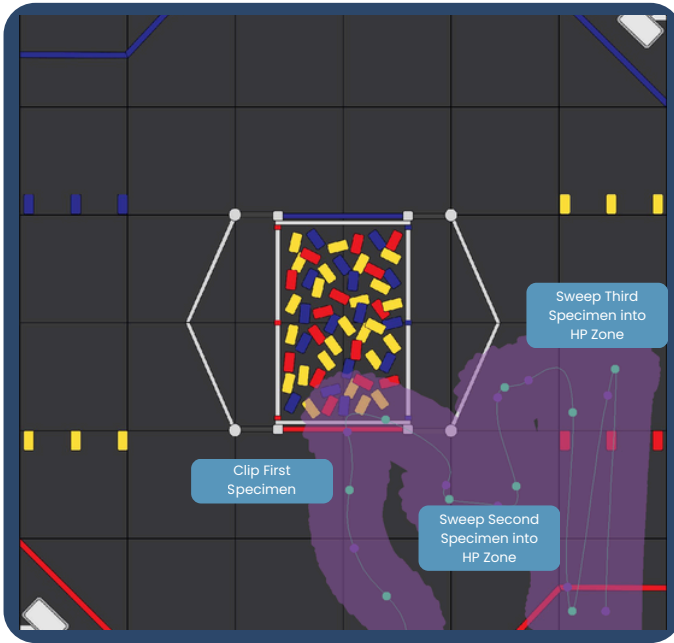
To relieve this, we switched to using the path visualization tool **RRPathGen** to automatically calculate lengths and heading angles.

The **transition** to RRPathGen allowed us to accomplish a **60-point 3-specimen** autonomous without a second preloaded specimen from the human player. This greatly **differed** from the 3 specimen autonomous we accomplished in Meet 3 as it was **reliant on our teammates pre loading a specimen**, and if they did we could only accomplish a **2-specimen or 40-point auton**.

Through Roadrunner, we maximized the potential of our robot and achieved the **highest autonomous score** we have ever accomplished.

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{d}{dt} e(t)$$

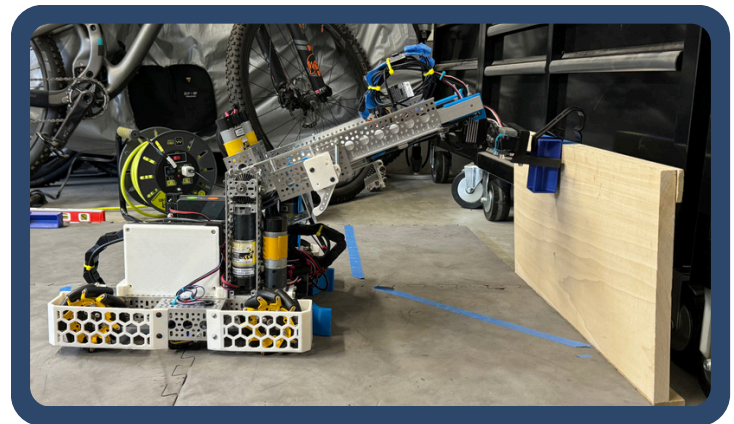
We also learned the Proportional-Integral-Derivative (PID) formula to see how RR works internally!



Our RRPathGen Splines

POST ILT APPROACH

Going into ILT, we encountered many issues involving **inconsistencies with the human player** placing specimens for the robot to grab. This often led to **penalties** and resulted in some wasted unscored specimens. To account for this, we changed our autonomous to grab from a **consistent spot on the wall**, minimizing any inconsistencies via our wrist (seen on the right)



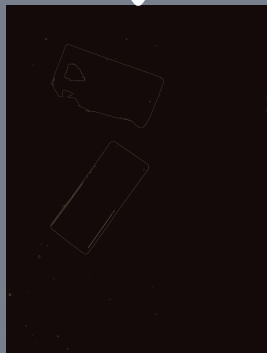
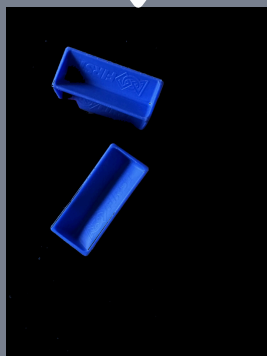
OUR RR STRATEGY

Introducing Roadrunner completely shifted our game strategy going into Meet 3 and ILT. After clipping **three specimens** onto the submersible in **autonomous**, we proceed by:

1. **Quickly collecting the opposing teams yellow samples** to prevent them from scoring
2. Using horizontal extension to grab red/blue samples from inside the submersible, **taking them to HP zone, and clipping onto submersible**
3. Placing our yellow samples in high basket

COMPUTER VISION (OPENCV)

PROCESS



*Blue sample detection shown, algorithm also supports red sample detection

Grabbing from the submersible was difficult with the claw alone; our solution was to implement an **auto-align feature** with **OpenCV** and **machine learning**.

STEP BY STEP

Align the claw with the center of the robot and click the auto-align button (Y)

Wait 300 ms for stability and rotate the servo to the **altered position**

Press **X** to start the CV process, which captures a frame from the **Logitech** camera and converts from **RGB to HSV** for easier use

Apply a red/blue **color mask** that filters pixels through a set color range, removing pixels which don't meet the threshold.

Apply a **Gaussian blur** and the **Suzuki Abe algorithm** to simplify edge detection and capture contour vectors

The angle is then **normalized** to remove negative angles and adapt for servo positions:

$$0.5 \pm \frac{(90 - \text{angle})}{360}$$

Obtain the two furthest vector points and plug them into our **angle calculation equation**

OBSTACLES

- Trade off analysis: We initially used **Canny Edge Detectors**, but they proved **weaker for edge detection** due to background noise detection.
- We initially did not have a **color mask**, but it posed many problems that were fixed through their addition
- The algorithm initially detected **all edges**, but we narrowed it down to the **longest diagonal edge** for improved accuracy



Camera that was disassembled + used:
Logitech C920x Camera

EQUATIONS

GAUSSIAN BLUR

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

SUZUKI ABE ALGORITHM

$$P_{i+1} = P_i + (dx, dy)$$

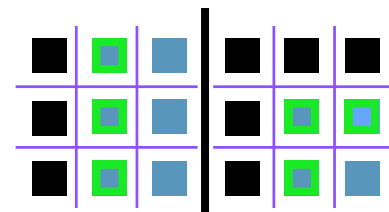
ANGLE CALCULATION

$$\theta = \frac{180}{\pi} \tan^{-1} \left(\frac{y_2 - y_1}{x_2 - x_1} \right)$$

CONTOURS

$$C = \{P_1, P_2, P_3, \dots\}$$

Contours are vectors with multiple points. The **constant C** declares the intensity for which a contour can operate on. Through an **8 pixel detection**, contours detect points which are likely to be edges (seen below)



INTRODUCTORY PROGRAM



To improve the skills of entering team members, we chose to put new members through a one-year **Introductory Program**. We trained any aspiring team member through our **comprehensive curriculum**. We taught all members valuable coding and building skills to ensure they're capable for a future on the team. This year, the program has **greatly expanded**, a strong sign for our club's performance in future seasons.

+148%

INCREASE IN MEMBERSHIP
INTEREST FROM 2024

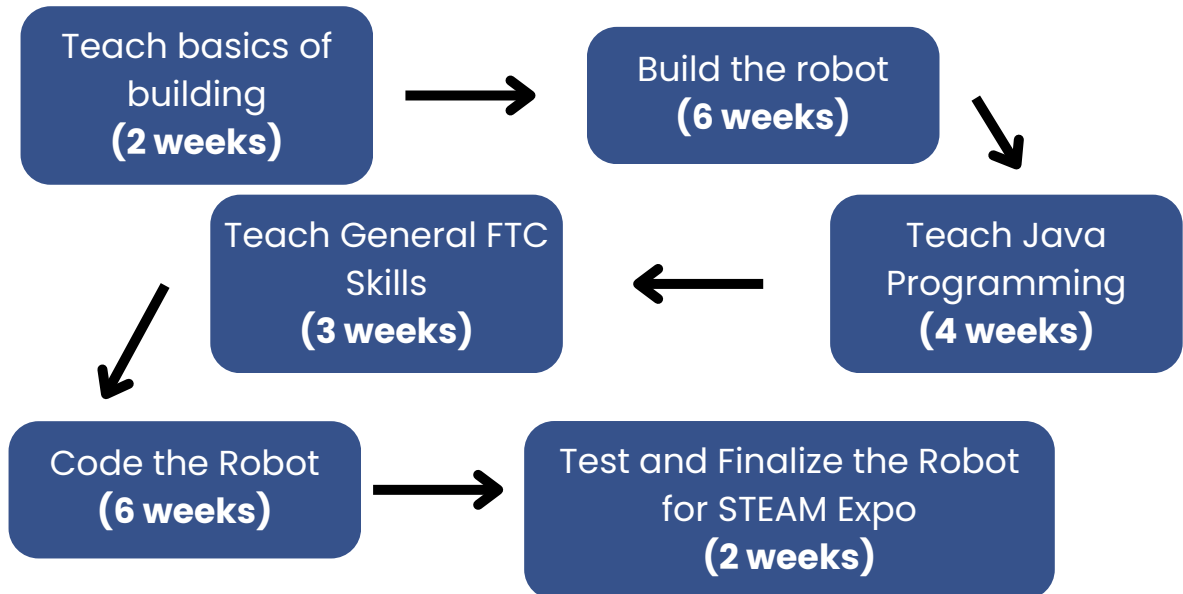
+128%

INCREASE IN ACTIVE
MEMBERSHIP FROM 2024

~ 46%

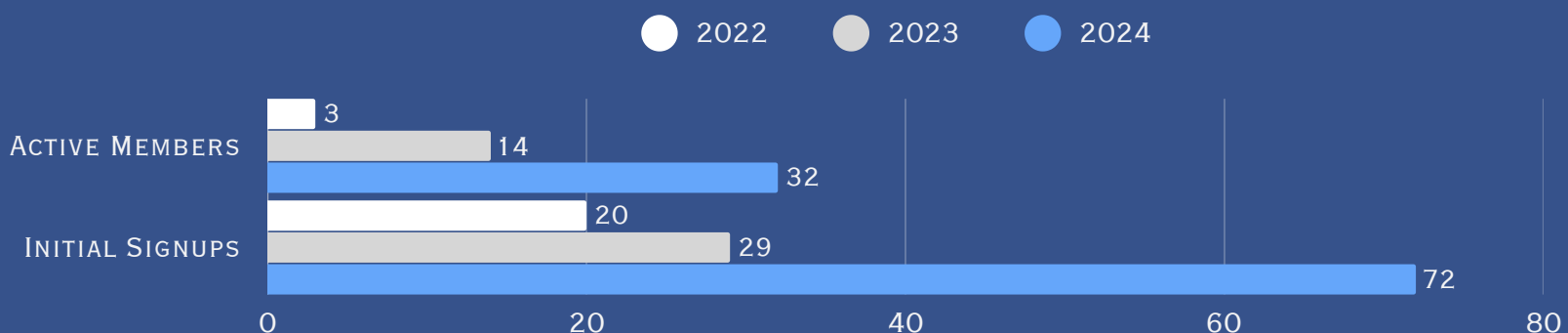
CONSISTENT RETENTION
RATE AMONG 1ST YEARS

OUR CURRICULUM



2024 -2025 Improvements

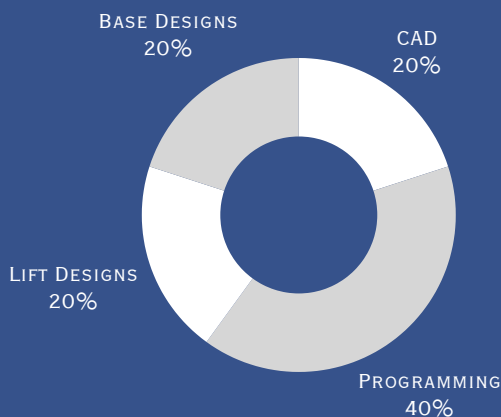
- Increased amount of Student Teachers to improve engagement
- Improved Curriculum (overhauled of coding slides, taught Misumi slides)
- Purchased more kits to ensure more members are involved



SUMMER WORKSHOPS

To address a difficult transition from the Intro Program into the team environment, we hosted Summer Workshops for incoming team members. We supplemented their previous experience with competition-like challenges and **taught advanced skills** like computer vision, CAD, and lift mechanisms.

Outside of the lessons, **members were able to form bonds** before the start of the season, improving our synergy in the future.



These workshops have provided perfect opportunities for **knowledge transfer**, such as:

- Two experienced members **creating interactive CAD videos (seen above)** that will teach incoming members the basics for years. Just this year they have resulted in the creation of various CAD designs.
- **Graduating students getting chances to pass down niche advice**, from placement of motors to ensure hex-driver accessibility to proper use of Servo programmers.

SUMMARY

Through both the Intro Program and Summer Workshops, we have **tremendously improved the retention, performance, and skills of our team**. More importantly, our establishment of these programs will continue to help us **consolidate our experience as a club for years onward**.

COMMUNITY ENGAGEMENT

Apart from building the team up internally, we have **actively sought opportunities for us to engage within our community** and form beneficial bonds: from individual engineers, to other schools, to universities, to larger organizations.

+5

CONTINUING COMMUNITY
PROGRAMS

+2

NOVEL CONNECTIONS
ESTABLISHED

COLLABORATIONS

ARMY

We entered into collaboration with the local US ARMY recruiter office, establishing a bond which we hope to continue for years to come. Additionally, our connection has brought more opportunities to all of us:

- **For 20381:** We learned more about the ARMY-specific engineering programs and gained the resources/knowledge to build similarly complex robots
- **For the ARMY:** They have gained a future partner to continue to **encourage STEM within the younger generations**



SPACE VECTOR

We also continued to work with Space Vector Corporation, **learning from their engineers** to **utilize encoders correctly** and **complex linear mechanisms** which allowed us to excel within our scrimmages.

FUNDING USE

- Misumi Slides - *Upgraded from Makeshift GoRails and allowing precision/complexity*
- Odometry Pods - *Enabling RoadRunner implementation*
- 3D Filament - *Supporting each iteration of each Custom Part*

\$5000+

FUNDS RAISED FROM
DONATIONS

65+

STUDENTS REACHED
THROUGH ARMY PROGRAM

HILLSIDE MIDDLE

Our team helped **bring ambitious ideas to life through hands-on robotics challenges**. Rather than assembling the same Makeblock kits, we **introduced new challenges and concepts** ranging from CAD, utilizing gear ratios, all while promoting curiosity.

40+

MIDDLE SCHOOL STUDENTS
MENTORED

PEPPERDINE UNIV.

We attended a **Star Wars themed Pepperdine Faculty Party** where we showcased our FTC program and robot. Not only did this experience allow us to exchange insights and enthusiasm about STEM Education, **but we also shared our nerdy love of Star Wars**. Our two communities engaged in more meaningful ways: **bonding over Driver Control testing and Karaoke over Steak Dinner**.



15+

SERVINGS OF TRI-TIP
DEVoured

30+

FACULTY MEMBERS
CONVERSED WITH

FTC PROMOTION

STEAM EXPO + HOMESHOW

We invited team 6282 Simi Valley Robotics to take central roles within our local community events. At both the STEAM Expo, and SSHS's Homeshow, which were open to the public, we **simulated FTC matches** for younger students to encourage entering our program. Not only did we engage with and learn from local engineers, we also **inspired the next generation of local youth** and our Introductory Students. We allowed them to lead our event to **embody the quick-thinking mindsets** needed in the coming years.



90%

INTRO-STUDENT
INVOLVEMENT

450+

STUDENTS/PARENTS
REACHED

7+

LOCAL SCHOOLS
INVOLVED

SCALE21x LINUX EXPO



We held a booth at the SCALE21x event, **inspiring countless young engineers** and speaking with professionals of many ages. We used the day to **explain the FTC competition** and even worked with STEM-involved parents to **teach them how to bring FIRST to their school**. Additionally, we showcased a **1:4 scale model of the Mars Rover** with a JPL engineer, seen above.

2024-2025 Plans

- Continuing JPL Joint Program, Building Functional 1:4 Mars-Rover
- Inviting 3+ local teams for simulated FTC matches
- Full-Day CAD Workshops for estimated 100+ Students

1500+

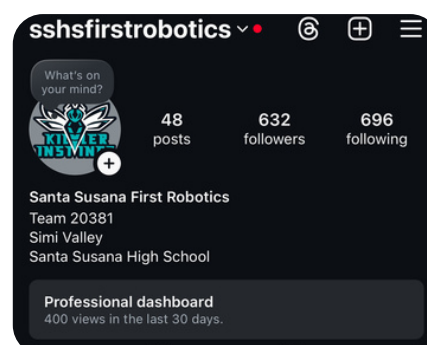
PEOPLE REACHED

10+

COMPANIES NETWORKED
WITH

INSTAGRAM

- **Increased Engagement:** Over 5,000+ views per post and 200+ new followers gained.
- **Promoting FIRST Values:** Advocated STEM, innovation, and teamwork.
- **Community Outreach:** Expanded team visibility and local support.



5000+

INSTAGRAM VIEWS